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New graphical method to measure silent periods evoked by transcranial magnetic stimulation

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Abstract

Objectives: Manual methods of measuring duration of cortical silent periods (CSP) evoked by transcranial magnetic stimulation (TMS) depend upon subjective visual estimation of onset and offset. Because of this, the measurements are susceptible to poor rater reliability. We describe a graphical method to measure silent periods with greater precision. The statistical process underlying this new method is simple and particularly suited to signal detection in serially dependent data.

Methods: TMS-evoked silent periods were recorded in 13 healthy subjects. Two investigators subjectively measured silent period duration on each subject to estimate rater reliability. Using the graphical method, the mean and 99.76% variation limits of pre-stimulus electromyogram (EMG) activity were computed. Each averaged trial was displayed and CSP onset and offset detected when post-stimulus EMG activity moved outside the 99.76% limits.

Results: Maximum variation in silent period duration was 21.8 ms between the two investigators' subjective measurements. Silent period duration measured with the graphical method closely approximated measurements obtained using the manual method. It was possible to automate the procedure.

Conclusions: This graphical method allowed precise measurement of CSP duration, independent of subjective estimations of onset or offset points. Further studies are necessary to determine if this method can provide a framework for other physiologic measures. © 2001 Elsevier Science Ireland Ltd. All rights reserved.

Keywords: Cortical silent period; Graphics; Statistical process control; Transcranial magnetic stimulation; Physiologic measurement

1. Introduction

The cortical silent period (CSP) refers to a silence in the electromyogram (EMG) of a voluntarily contracting muscle elicited by transcranial magnetic stimulation (TMS) (for review, see Hallett, (1995)). The duration of the CSP at a given stimulus intensity reflects the integrity and excitability of cortical inhibitory mechanisms, some of which are probably mediated through gamma-aminobutyric acid-B receptors (Werhahn et al., 1999). At maximum stimulator output, the duration of the CSP in a hand muscle of normal adult subjects is typically in the range of 150–300 ms (Cantello et al., 1992).

Although the concept underlying the CSP is widely accepted, investigators use different definitions for onset and offset. Variation in CSP definitions has a direct effect on duration. A selection of reports studying CSP duration in patients with Parkinson disease can serve to illustrate the difficulty that arises between different investigator groups (Berardelli et al., 1996; Cantello et al., 1991; Priori et al., 1994; Ridding et al., 1995; Young et al., 1997). These studies used subjective methods to visually estimate onset and offset of the CSP in averaged trials from individual patients. In all but one study, CSP onset was measured from the end of the motor evoked potential (MEP). The remaining study measured onset from the point of stimulation. Offset was defined as either the first appearance of EMG activity after CSP onset or the point when the post-stimulus EMG reached the average level of EMG activity prior to the stimulus. This average was not statistically defined, but was visually estimated by the investigator. Not surprisingly, mean CSP duration in normal controls in these studies varied from 164 to 217 ms even when stimulus intensity and level of voluntary muscle activation were comparable.

Rater-reliability adds a further confounding factor when

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manual methods are used to measure CSP duration. For example, in one study (Uncini et al., 1991) CSP duration was determined "by visual inspection of the rectified EMG as that portion of the tracing clearly lower in than amplitude than any other part of the tracing beforehand or afterwards" (Uncini et al., 1991). Using a similar visually guided method, Nilsson et al. (1997) found that CSP duration varied by as much as 20 ms when two investigators from the same group measured the duration.

Problems such as those described above affect both the internal and external validity of a study. In an attempt to standardize the measurement of CSP duration, a computer-aided statistical analysis of CSP duration was proposed (Nilsson et al., 1997). The method identifies CSP onset and offset using unpaired Student's *t* tests. A 'sled' moves along overlapping 2–4 ms epochs of post-stimulus EMG and compares each epoch with the mean pre-stimulus EMG. The onset of CSP is defined as the point at which the post-stimulus EMG is significantly different from the pre-stimulus EMG. The first point at which the *t* test fails to detect a significant difference is defined as CSP offset (Nilsson et al., 1997).

The paired Student's *t* test is commonly used in neurophysiology for signal detection, particularly in the analysis of evoked potentials (Guthrie and Buchwald, 1991). However, studies have shown that it may not be the most sensitive method of signal detection (Achim, 1995). Of greater concern, however, is that Nilsson's modification uses the unpaired Student's *t* test to compare pre- and post-stimulus EMG data points. Since these data observations are not independent of one another, this violates the statistical assumptions and conditions for which the unpaired *t* test is intended. Furthermore, the method requires a large number of *t* tests. For example, after CSP onset the method calls for a *t* test every 2 ms until the *t* test is significant at a level of $P < 0.005$ and thereafter every 0.1 ms until the *t* test is no longer significant at a level of $P < 0.05$. In the data presented in the report, duration of EMG suppression is approximately 38.6 ms. Thus the data could call for as many as 380 *t* tests depending on the level of significance at each stage. The authors fail to discuss the implications this could have on the interpretation of the results, viz. the need to adjust for multiple comparisons. In addition, the need for multiple *t* tests makes the method somewhat cumbersome. Although the authors point out that computer programs can be written to perform these *t* tests, this may not be available to all those interested in measuring CSP duration.

The present report proposes the use of a simple graphical method to measure cortical inhibitory phenomena that can add precision to manual methods. Unlike the Nilsson method, it does not use multiple statistical tests, but is based solely upon the simplest of arithmetic calculations: calculation of the average pre-stimulus EMG activity level and the mean consecutive difference (MCD) between pre-stimulus EMG data points. A detailed example of the MCD is given in Appendix A. From these, the 99.7% limits of variation are derived. These limits act as a guide to precisely determine CSP onset and offset by observing how the post-stimulus EMG, plotted on a simple graph, differs in variation from the pre-stimulus EMG. The method described here is particularly suited for serial data collected from the same patient, overcoming the statistical disadvantages of the more common inferential statistical methods (Student's unpaired *t* test and analysis of variance). It should have considerable merit for EMG silent period detection.

2. Subjects and methods

Eight healthy children (5 female) with a mean age of 12.2 years (SD 0.98; range 10.5–13.6) were recruited for a study determining the duration of the CSP in children. Five healthy adults (one female) with a mean age of 33.5 years (SD 9.5; range 23–47) were also recruited to provide an adult control group. All subjects (and their parents where applicable) gave written assent/consent to the study, which was performed with the approval of the local Institutional Review Board Committee. Subjects were free of any neurological or psychiatric illness and all wore hearing protection (earplugs) for the duration of the testing session. Bipolar EMG was recorded from surface electrodes positioned over the right first dorsal interosseus (FDI) muscle and distally over the corresponding index finger (distal tendon reference).

Subjects were seated in a comfortable chair, with a firm pillow on their lap. Activation of right FDI was achieved by asking the subjects to squeeze a firm rubber object as hard as they could, using the thumb and index finger. Level of muscle activation was monitored with a real-time oscilloscope and auditory feedback. A Magstim 200 magnetic stimulator (The Magstim Company Ltd, Spring Gardens, Whitland, Carmarthenshire, UK) and a focal figure of 8 coil (diameter of each wing 70 mm) were used to stimulate the left motor cortex. The coil was placed tangential to the scalp with the handle pointing backwards and 45° away from the midline (Ziemann et al., 1997). Once the optimum site for eliciting motor responses in the right FDI was identified, resting motor thresholds (RMT) and active motor thresholds (AMT) were determined to the nearest 1% of the maximum stimulator output. RMT was defined as the minimum stimulus intensity which elicited small MEPs > 50 μV in at least 5 out of 10 consecutive trials. AMT was obtained during maximal muscle contraction and was defined as the minimum stimulus intensity which elicited a MEP > 100 μV in the average of 5 consecutive trials.

The CSP was recorded in all subjects using a stimulus intensity of 1.5 times the AMT. Each trial collected EMG activity for a total of 600 ms, with a 100 ms pre-stimulus window. EMG activity was sampled at a 2 kHz analog-to-digital conversion rate using Spike2 for Windows software and the micro1400 unit (a data acquisition interface with

high-speed waveform capture, digital event I/O, stimulus sequencing and on-line analysis capabilities) (Cambridge Electronic Design (CED), Cambridge, UK). Ten individual trials with acceptable levels of EMG activity were rectified and averaged. The averaged trial was displayed for manual measurement and subsequently exported in ASCII format and imported to a Microsoft Excel® spreadsheet.

The statistical process control method establishes the outer variation limits of the pre-stimulus EMG based on the mean successive difference between individual data points (see Appendix A). Smaller successive differences will bring the variation limits closer to the mean; larger successive differences will move the limits farther from the mean. Since onset and offset times are identified from these MCD derived limits, MEP and CSP duration will vary depending on the magnitude of successive differences. In addition to biological sources of EMG variability, the following data acquisition and analysis factors may also affect the magnitude of successive differences: (1) sampling rates; (2) level of EMG activation; and (3) number of trials included in the average.

Sampling rate directly affects the successive difference between two data points of EMG (Gitter and Stolov, 1995). Successive differences between data points occurring at frequent intervals (faster sampling rates) will be smaller than successive differences between data points at less frequent intervals (slower sampling rates). To determine how changes in sampling rate might affect this graphical method, we recorded the CSP at 10 kHz in one subject. We then calculated the MCD 3 times on the average of 10 trials using successive differences between data points at 0.1 ms intervals (10 kHz), 0.2 ms intervals (5 kHz), and 0.5 ms intervals (2 kHz). Variation limits were derived from each of these MCD and CSP duration was measured 3 times using the different variation limits.

The coefficient of variation of EMG signals in different trials obtained from the same individual is 21%, even when experimental conditions are kept exactly the same (Araujo et al., 2000). This variability increases if the subject is unable to sustain steady muscle contraction. This may occur more commonly in children than in adults. Increased variation of EMG data may increase successive differences between data points. To determine the effect of increased EMG variability, we compared the MCD obtained from EMG data in the children with the MCD from the adults. In addition, we asked 4 of the 5 adult subjects to simulate variable EMG by instructing them to alternately release and contract their muscles while evoking the CSP. Stimuli were given randomly during these repetitive contractions so that some were delivered when the muscle was relaxed and others were given at various stages of contraction.

Averaging reduces the differences between data points. This will decrease successive differences between data points, but may also alter the appearance of the data. Averaging could, therefore, affect this method of measuring CSP duration in two ways: by reducing the MCD and by altering the appearance of the EMG at the onset and offset points. To determine the effect of averaging, 50 trials were collected in all adult subjects. We measured the MCD and CSP duration from the averages of 10, 20, 30, 40, and 50 trials. CSP duration was also measured on the first and last 10 trials collected. In this way, we could determine if other, perhaps biological, factors altered CSP duration when evoked multiple times, independent of the averaging process.

3. Data analysis

3.1. Definition of CSP onset and offset

For both the subjective and objective methods, CSP onset was defined using the two most prevalent definitions: (1) from the onset of the MEP and (2) after the end of the MEP from the start of the silent period itself. Offset occurred with a return of sustained post-stimulus EMG activity. CSP measurements incorporating the preceding MEP were labeled 'CSP-M' while those measuring the silent period proper were labeled 'CSP'.

3.1.1. Subjective method

Manual measurements of CSP duration elicited during steady EMG contraction were made by the first author on the average of 10 trials for all subjects. After data collection, the rectified averaged trial from each subject was displayed and vertical cursors were placed at the points estimated to represent silent period onset and offset according to the above definitions. Spike2 software measured the exact time interval between the cursors to calculate silent period duration.

For the purposes of a reliability exercise, CSP duration was measured by a second investigator (child data = DAB; adult data = CAB) in the data from each subject after reaching consensus with the first author on the decision criteria used to determine CSP onset and offset as outlined above.

3.1.2. Objective graphical method

The additional requirements for this simple procedure were only a spreadsheet with graph making capabilities. Each rectified averaged trial was exported, in ASCII format, to a spreadsheet. Using statistical process control (Shewhart, 1986; Wheeler, 1993), variation limits were determined using the MCD of data points in the pre-stimulus EMG (start of data collection to 2 ms before the time of the stimulus (100 ms window)). These limits covered 99.76% possible pre-stimulus EMG data points (equivalent to 3 standard deviations (SD)) (Pfadt and Wheeler, 1995; Wheeler, 1993) (see Appendix A). After the entire trial was displayed, horizontal cursors were placed representing the mean pre-stimulus EMG level and upper and lower variation limits. Onset of CSP-M was defined as the onset of the MEP. Onset and offset of the silent period proper were identified using the following criteria: (1) time of onset was the first of 5

consecutive data points to fall below the lower variation limit; (2) all subsequent data points were considered part of the CSP until there was a return of sustained EMG activity; (3) time of offset was defined as the first datum point that fell above the lower variation limit if 50% or more of the data points in the following 5 ms window were also above the lower variation limit.

3.1.3. Reliability

Manual and graphical measurement of CSP duration from each individual were compared using intraclass correlation (ICC) (Bartko and Carpenter, 1976). Rater reliability for both CSP-M and CSP was also calculated using ICC and a paired Student's *t* test compared the means obtained by the two investigators.

3.1.4. Effect of EMG variability

An unpaired *t* test compared the MCD calculated from the pre-stimulus EMG in children with that in the adult subjects. A paired *t* test compared the MCD calculated from the trials during steady EMG activity with those during simulated variable activity from the same individuals. There is some controversy regarding the effect of muscle activation on CSP duration (Inghilleri et al., 1993; Wilson et al., 1993). For this reason, we did not compare CSP duration between the two EMG activation levels.

3.1.5. Number of trials averaged

A paired *t* test compared CSP duration in the first and last 10 trials recorded. A repeated measures ANOVA was performed with MCD as the dependent variable and number of averaged trials (10 vs 20 vs 30 vs 40 vs 50) as the within group variables. A second repeated measures ANOVA was performed with CSP duration as the dependent variable and number of averaged trials (10 vs 20 vs 30 vs 40 vs 50) and method of measurement (manual vs graphical) as the two within group variables. We expected to find that the MCD would decrease with increasing number of trials included in the average and that an increase in the CSP duration would be apparent regardless of the method used to measure the duration. All reported values are mean and SD unless otherwise stated.

4. Results

All subjects completed the TMS session without difficulty. RMT levels were 52% (SD 5; range: 45–63) for the children and 40% (SD 13; range: 34–51) for the adult subjects. AMT levels were 31% (SD 4; range: 25–40) for the children and 30% (SD 6; range: 25–38) for the adult subjects. CSP-M and CSP measurements comparing the two methods are shown in Table 1. Although measurements using the graphical method closely approximated the manual measurements (ICC : CSP-M = 0.96; CSP = 0.94), the difference between graphical and manual measurements was greater in the child subjects than in the adult subjects.

4.1. Rater reliability

ICC of CSP-M and CSP measurements for the two investigators was good (child subjects: CSP-M = 0.78; CSP = 0.89; adult subjects: CSP-M = 0.97; CSP = 0.99) and a paired *t* test revealed no difference between the two investigator CSP-M or CSP means. The maximum differences were as high as 21.8 ms for CSP-M and 18.5 ms for CSP. Only one was exactly the same duration and 3 (23%) had a difference of less than 1 ms.

4.2. Effect of sampling rate

Fig. 1 and Table 2A show the variation limits derived from the MCD calculated from successive differences between data points occurring every 0.1 ms (10 kHz), 0.2 ms (5 kHz), and 0.5 ms (2 kHz). There was a roughly linear relationship between the sampling rate (frequency of data points) and the MCD. The figure shows how the variation limits change as the sampling rate increases. Because the MCD variation limits derived from data with 5 and 10 kHz sampling rates are closer to the mean, CSP duration increases.

4.3. Effect of variable EMG

Unpaired *t* test showed no difference between the children's pre-stimulus MCD (mean 0.031; SD 0.02; range 0.009–0.057) and the adult subjects' MCD (mean 0.025; SD 0.01; range 0.011–0.025). The MCD in the simulated variable EMG was larger than that seen in the steady EMG (mean MCD: steady = 0.02 (SD 0.01; range: 0.01–0.03); variable = 0.03 (SD 0.01; range: 0.02–0.04)) but this did not reach significance on a paired *t* test.

4.4. Effect of averaging

Table 2B shows the mean CSP duration for both methods of measurement when increasing number of trials are added to the average. The mean difference in CSP duration between the first and last 10 trials recorded was −2.1 ms (SD 8; range −14 to 5.5; paired *t* test; $P = 0.58$). The *F* test, MCD by number of trials, was significant (MCD decreased

Table 1
CSP duration in children and adults. Comparison between manual and graphical methods of measurement

	Method of measurement			
	Children Duration (ms) mean ± SD		Adults Duration (ms) mean ± SD	
	Manual	Graphical	Manual	Graphical
CSP-M	42.2 ± 23.1	53.8 ± 34.5	91.3 ± 57.4	90.2 ± 56.4
CSP	25.5 ± 22.9	32.5 ± 29.0	68.8 ± 49.8	66.8 ± 59.8

with increasing number of trials averaged) ($F = 8.68$; d.f. 4,24; $P = 0.04$; Huynh–Feldt adjustment). Post-hoc analysis showed this difference to be between the MCD calculated from 10 averaged trials and the MCD calculated from 30, 40, and 50 averaged trials.

In the two factor repeated measures ANOVA, the 3 F tests, number of trials ($F = 3.34$; d.f. 4,49; $P = 0.11$); method ($F = 1.66$; d.f. 1,49; $P = 0.26$) and the interaction of number of trials by method ($F = 4.54$; d.f. 4,49; $P = 0.11$; Huynh–Feldt adjustment), were all statistically non-significant. This indicated that although manual and graphical measurements of CSP duration lengthened as more trials were added to the average, this increase was not significant.

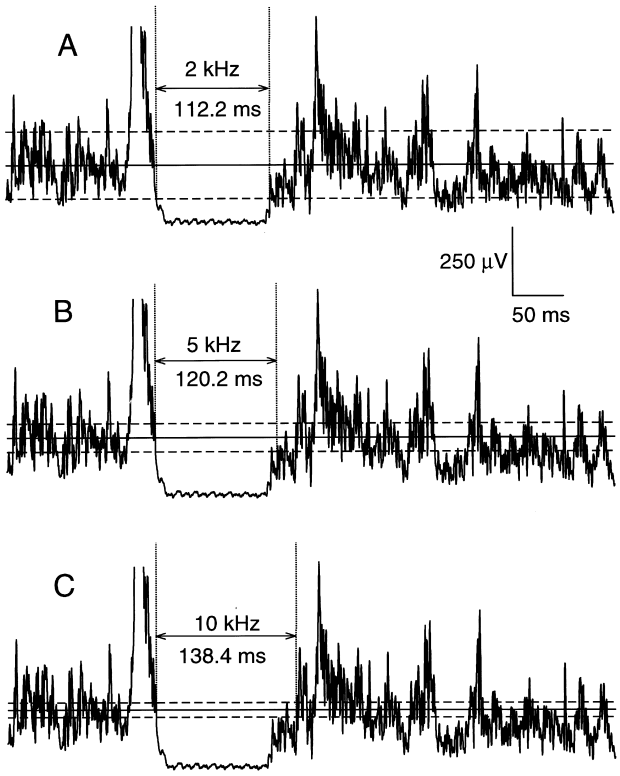

Fig. 1. Effect of sampling rate on CSP duration. The 3 tracings are the same average of 10 rectified trials recorded from the right FDI muscle showing a silent period evoked by a stimulus 1.5 times AMT applied to the left motor cortex. The sampling rate was 10 kHz. The silent period is measured from the onset of EMG suppression (CSP). The MCD was calculated 3 times using successive differences between data points occurring every 0.1 ms (equivalent to 10 kHz), 0.2 ms (equivalent to 5 kHz), and 0.5 ms (equivalent to 2 kHz). Variation limits were derived from each of these MCD. For each tracing, horizontal lines indicate the pre-stimulus EMG mean (solid line) and upper and lower variation limits (dashed lines). Vertical dotted lines indicate CSP onset and offset. Note that when MCD is calculated using every data point (equivalent to 10 kHz), the variation limits are closer to the mean. Time of offset was defined as the first datum point that falls above the lower variation limit if 50% or more of the data points in the following 5 ms window are also above the lower variation limit. With narrower limits, silent period offset is detected later, increasing silent period duration. The variation limits derived from the 2 kHz MCD best approximates visual estimation of CSP offset.

5. Discussion

Despite its seeming simplicity, visual inspection of graphic data is a complex and subtle process (Fisch, 1998). In keeping with earlier reports (Nilsson et al., 1997), we found differences of up to 22 ms for the CSP when two investigators used a visually guided manual method to measure CSP duration from the same subjects. Although small, a difference of this magnitude may be critical to differentiate healthy controls from patients as has been previously reported (Ziemann et al., 1997). In addition, the subjective nature of measuring by visual inspection can seriously compromise the ability to replicate findings in similar patient populations. While results from individual studies may give limited insights into the physiological nature of a disorder, little definitive knowledge can be gained unless it is possible for the results to be replicated. Other difficulties affecting replicability of studies, such as the stimulus intensity used to elicit CSPs, the degree of muscle activation, and definitions of onset and offset can be overcome by establishing a universal standard among researchers. Rater reliability is less amenable to such a solution.

The ICCs were smaller for the children's data compared to the adults, suggesting that it was harder to find consensus regarding onset and offset in these subjects. The greater difference between graphical and manual methods in these young subjects indicates that this lack of consensus was not solely due to poor rater reliability, but was related to the difficulty in interpreting the EMG tracings. MEPs are more complex in children (Jones et al., 1996) and we observed that silent period offset in the child subjects occurred with a less abrupt return to pre-stimulus EMG activity. Rater reliability may be of greater concern in this population, suggesting a more critical need for methods that add precision to CSP duration measurements.

The method proposed in this report may also allow greater precision with other atypical patterns. For instance, it may be difficult to decide whether EMG activity occurring during the silent period are breakthrough waves or if the silent period is divided. In these situations, the use of the variation limits would be helpful to determine whether the EMG activity meets criteria for a true offset without relying on the subjective decision of an investigator. Other atypical patterns may also occur in children (Garvey, unpublished material). For example, young children may show a resumption of EMG activity after the MEP, but before the silent period begins. These unusual patterns could be precisely quantified using this method and may be helpful to determine the effect of development on the silent period.

The present method is based on statistical process control. This method was first proposed in the 1930s by Shewart (1986). It allows observation of time-oriented data and assessment of patterns of variation and the reliability and validity of this method has been rigorously tested in the manufacturing and service industries (Wheeler, 1994). It is particularly suited for signal detection in serial data

Table 2
Factors affecting the MCD

(A) *Effect of sampling rate on calculation of the MCD*

Sampling rate (kHz)	MCD[a]	Duration (ms)	
		CSP-M[b]	CSP[c]
2	0.046	138.6	112.2
5	0.019	145.6	120.2
10	0.010	164.0	138.4

(B) *Effect of averaging on silent period duration as measured by the graphical method*

Number of trials averaged	Duration (ms) ± SD			
	CSP-M[b]		CSP[c]	
	Manual	Graphical	Manual	Graphical
10	91.3 ± 56.4	90.2 ± 56.5	68.8 ± 51.2	66.8 ± 49.7
20	92.5 ± 57.7	93.1 ± 57.6	70.4 ± 52.6	70.3 ± 50.6
30	94.6 ± 56.3	95.2 ± 56.3	73.2 ± 51.2	73.0 ± 49.7
40	92.7 ± 58.8	99.1 ± 58.8	71.0 ± 52.4	77.0 ± 52.1
50	97.4 ± 59.1	98.9 ± 59.1	75.7 ± 54.3	77.1 ± 52.4

[a] MCD, mean consecutive difference.
[b] CSP-M, cortical silent period measured from the onset of the MEP.
[c] CSP, cortical silent period measured from the end of the MEP.

from the same subject portrayed in graphical form, thus overcoming the statistical problems inherent in Nilsson's method. Statistical process control has recently been introduced to aid in the analysis of clinical data (Boggs et al., 1999; Pfadt and Wheeler, 1995; Pfadt et al., 1992). To our knowledge, this is the first time this procedure has been applied to neurophysiologic data.

In any signal recognition task, two mistakes are possible: the false positive signal (interpreting random variation as a meaningful departure from the pre-stimulus EMG) and the false negative signal (failing to detect a signal when it is present). The advantage of this method is that standardized decision rules or criteria can be established to minimize these errors (Pfadt and Wheeler, 1995). These criteria must encompass the variation limits and a 'run' pattern, and might also include duration criteria (for example, 10 ms minimum duration for EMG suppression to be considered a true silent period). In this report, the variation limits were set to 99.76% of the data. Since random data points fall outside 99.76% variation limits less then 1% of the time, a run of 5 consecutive post-stimulus EMG data points outside these limits can safely be considered different from the pre-stimulus mean (Pfadt and Wheeler, 1995). Duration criteria were considered for this study but discarded given the study population. CSP duration in both normal children and those with Tourette syndrome can be close to zero (Moll et al., 1999). Different decision criteria can be adopted for other study populations, depending on the level of confidence desired and the biological phenomena being measured. For instance, since EMG suppression in the ipsilateral silent period is usually less than in the CSP, narrower limits of variation could be used. The 99.76% limits in statistical process control are equivalent to 3 SD from the mean. The constant used to establish this limit is 2.66 (see Appendix A). The constants for 98.76% (2.5 SD), 95% (two SD), and 67% (one SD) variation limits are 2.22, 1.77, and 0.89, respectively.

Statistical process control establishes the limits of variation using the MCD. It would seem more straightforward to set the onset and offset limits based on the more conventional method of calculating the total SD of the raw pre-stimulus EMG data. However, the conventional method of calculating the SD assumes that each data point is independent of the others. Since EMG data is a set of serial dependent data points, calculating the SD violates this assumption. In addition, the SD is an insensitive signal detection guide (Fisch, 1998; Wheeler, 1993, 1994, 1995). To demonstrate this graphically, variation limits were calculated twice on one EMG tracing: using the conventional method of calculating the SD and and using the MCD (Fig. 2). Using the conventional SD limits, only the most profound EMG suppression is recognized as a silent period. This shortens the duration of both CSP-M and CSP.

Factors that affect the MCD may alter the accuracy of this method in detection of onset and offset points. We identified 3 such factors. The most important factor is sampling rate. Many different sampling rates are used for studies of CSP duration. For example, EMG data in this report was sampled

at 2 kHz; Nilsson et al. (1997) used a 10 kHz rate; Meyer et al. (1995) used a 5 kHz rate. Others do not report the sampling rate used during data collection. In this study, MCD decreased when successive differences were calculated from data points occurring at more frequent intervals. As the MCD decreased, the variation limits moved closer to the mean, increasing silent period duration. Thus, when an investigator wishes to replicate CSP measurements from a previous study using this method, she must calculate the MCD from consecutive differences in exactly the same way as the previous investigator. Although the simplest way to control this is by collecting data at the same analog-to-digital sampling rate, this may not always be possible. Alternately, she may opt to sample data at a higher sampling, as long as successive differences are calculated from data points occurring at the same intervals as in the previous study. For example, if the first set of data was collected at a 5 kHz sampling rate, and the second at 10 kHz, the MCD could be calculated from every other data point in the 10 kHz data (equivalent to 5 kHz intervals), effectively controlling for the sampling rate. In this study, graphical CSP measurements closely approximated manual measurements, when the MCD limits were calculated from data sampled at 2 kHz. Thus, for experimental conditions similar to the ones used here, we would suggest that investigators calculate successive difference from data points occurring every 0.5 ms (equivalent to 2 kHz sampling rate), regardless of the sampling rate used to collect the data. Investigators may choose to use faster intervals depending on the experimental conditions used in their study.

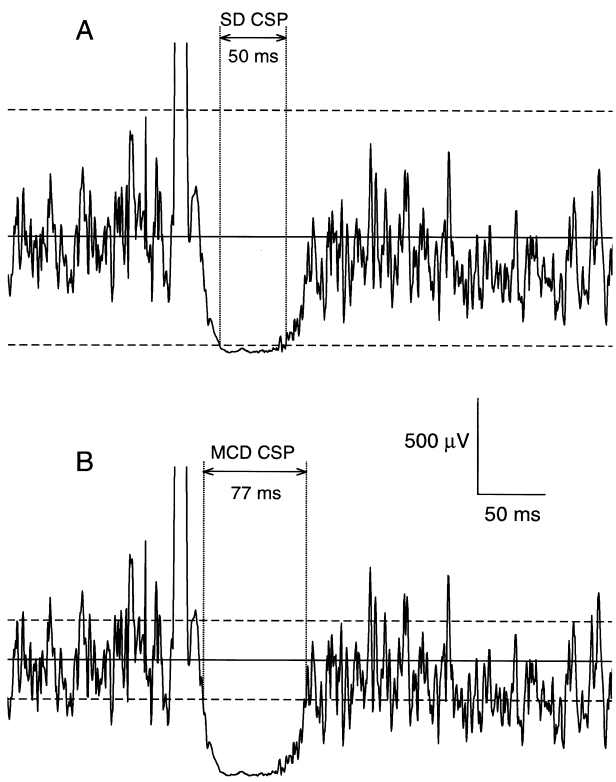

Fig. 2. (A,B): CSP measured using the conventional SD and the MCD. Both tracings are the same average of 10 rectified trials recorded from the right FDI muscle showing a silent period evoked by a stimulus 1.5 times AMT applied to the left motor cortex. The sampling rate was 2 kHz. For both the upper and lower tracings, horizontal solid line indicates the pre-stimulus EMG mean and the dashed lines represent the upper and lower variation limits. Vertical dotted lines indicate CSP onset and offset. Time of CSP onset was the first of 5 consecutive data points to fall below the lower variation limit. Time of offset was defined as the first datum point that fell above the lower variation limit if more than 50% of the data points in the following 5 ms window were also above the lower variation limit. In the upper tracing (A), variation of pre-stimulus EMG data has been estimated using the SD calculated in the conventional manner. The variation limit represents 3 SD above the mean (1.2909 mV); the lower limit is 2.5 SD limit (0.0717 mV) as the 3 SD limit was a negative value (−0.021 mV), that is, outside the possible range of rectified EMG data. Note that the lower 2.5 SD limit is only sensitive to the most extreme EMG suppression. In the lower tracing (B), the pre-stimulus variation has been calculated using the MCD. The dashed lines represent 99.76% variation above and below the pre-stimulus EMG mean. Note that the lower MCD limit detects an accurate representation of the silent period.

The MCD in steady EMG activity in adults was not significantly different from the MCD in children's EMG suggesting that this method can be used to compare silent period duration in children and adults. Because the MCD in simulated variable EMG was not significantly larger than that in steady EMG, it would be theoretically possible to use this method to reliably compare silent period duration in these two experimental conditions. However, it may be necessary to use different variation limits when doing this. In some of these trials, the lower 99.76% limits fell below zero because pre-stimulus mean EMG was low. This had not been a problem with the steady trials since these were recorded during maximal activation. Narrower limits of variation might also be useful when measuring CSP duration in trials with steady but minimal EMG activation.

Although the MCD decreased as more trials were included in the average, this did not significantly increase CSP duration. This suggests that this method is robust when between 10 and 50 trials are included on an average. However, we would recommend that the same number of trials be included in the average when replicating results from a prior study, to avoid any systematic bias inherent with the averaging process.

The signal detection method presented in this report offers a number of advantages to manual, visually guided, methods of measuring CSP duration. The method is much simpler than that proposed by Nilsson and is accessible to all investigators who can export data to a spreadsheet with graph making capabilities. The criteria allow precise identification of CSP onset and offset, and if specified, can be exactly replicated in later studies. Like Nilsson's method, it can be automated into a computer program reducing the time taken to analyze data. For these reasons, it should prove to be useful in standardizing the measurement of TMS evoked negative phenomena. Further studies will be required to confirm its usefulness and applicability for CSP measurement in patients with neurologic disorders.

Table 3
Statistical process control: calculating mean EMG activity level and the mean consecutive difference

Points	Time (ms)	EMG data	Difference ($D_n - D_{n-1}$)	Absolute difference ($\lvert D_n - D_{n-1} \rvert$)
1	0	0.435257		
2	0.5	0.401428	−0.033829	0.033829
3	1	0.342072	−0.059357	0.059357
4	1.5	0.323761	−0.018311	0.018311
5	2	0.373184	0.049423	0.049423
6	2.5	0.436691	0.063507	0.063507
7	3	0.434036	−0.002655	0.002655
8	3.5	0.450745	0.016708	0.016708
9	4	0.553909	0.103165	0.103165
10	4.5	0.687210	0.133301	0.133301
11	5	0.646576	−0.040634	0.040634
⋮	⋮	⋮	⋮	⋮
197	98	0.857971	0.009171	0.009171
198	98.5	0.695511	−0.162460	0.16246
199	99	0.668030	−0.027481	0.027481
200	99.5	0.715393	0.047363	0.047363
201	100	0.741165	0.025772	0.025772
Pre-stimulus EMG mean		0.634974		
MCD				0.077344

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Appendix A. Statistical process control: a precise method of measuring cortical silent period duration

In this graphical method of measuring silent period duration, signal detection is based on knowing the variation of the pre-stimulus EMG activity about its mean and establishing the limits of this variation. These limits can help distinguish random outlying data points from signals that represent real deviations from the mean. Variation of the pre-stimulus EMG data is established by forming subgroups of two data points. Successive differences between these data points are a measure of the variation of each subgroup. The average of all the subgroup variations (the mean consecutive difference (MCD)) estimates the overall variation of the pre-stimulus EMG data. This procedure is similar to that used in jitter analysis (Ekstedt et al., 1974). Applied to data in the form presented here, the method is commonly referred to as statistical process control (Wheeler, 1993). The process is illustrated in the following steps:

1. EMG variation is calculated only from the pre-stimulus time window. In the example below, the stimulus occurs at 102 ms. The data points are arranged with the number of each point in the first column, the time (ms) this point is observed in the second column and the raw EMG data in the third column ($D_1, D_2, D_3, \ldots D_{n-1}, D_n$). The fourth and fifth columns in the spreadsheet are mathematically derived values using very simple expressions The spreadsheet is presented in form of a table (Table A1).
2. In the fourth column, 'Difference' is the difference between successive data points ($D_1 - D_2$, $D_2 - D_3$, $D_3 - D_4$, … $D_{n-1} - D_n$) in pre-stimulus EMG. The purpose of this step is to form subgroups of size two so that the within-subgroup variation can be determined. Intrinsic variation computed in this fashion can reliably distinguish between signals and random noise (Wheeler, 1993).
3. In the fifth column, the absolute value for each consecutive difference is calculated. That is, all differences are taken as positive values.
4. The mean of the 201 EMG data points and the 200 absolute differences are calculated and appear at the bottom rows of the spreadsheet.
5. These steps are illustrated here using data from Fig. 2B. Data were sampled at 2 kHz and the stimulus occurred at 102 ms from the start of the recording.
6. Onset and offset limits to be applied to the graphical display of the data are then calculated from the MCD using the following formula:
   Mean pre-stimulus EMG ± [MCD × 2.66].
   The constant 2.66 comes from statistical tables appropriate for the average of the range of differences (Wheeler, 1993) and gives a measure of the 99.76% limits of variation of the pre-stimulus EMG data points. In this study, we chose to use the 99.76% variation limits, as they are equivalent to 3 SD. The constants for 98.76% (2.5 SD), 95% (two SD), and 67% (one SD) variation limits are 2.22, 1.77, and 0.89, respectively.
7. We then plot the data from the entire averaged trial on

an *X–Y* plot i.e. time by EMG plot. A horizontal line is drawn to represent the mean pre-stimulus EMG level. In our case, this is 0.635.

8. On the same plot, we draw the two horizontal lines corresponding to the lower and upper limits. In the example, the lower limit is $0.435[0.635 - (0.077 \times 2.66)]$ and the upper limit is $0.835[0.635 + (0.077 \times 2.66)]$.
9. Post-stimulus data starts 10 ms after the stimulus. While the horizontal lines representing the lower and upper limits give a visual guide to detect onset and offset of the silent period, precise time of onset and offset is determined by applying decision criteria in the raw EMG data. Time of CSP onset is defined as the first of 5 consecutive data points to fall below the lower variation limit. Time of offset is defined as the first datum point that falls above the lower variation limit if more than 50% of the data points in the following 5 ms window are also above the lower variation limit. This prevents false positive detection of CSP offset when a random datum point exceeds the lower variation limit, unless it is followed by a return of sustained EMG activity. It also takes into account the natural variability of EMG activity about the mean. Fig. 2B graphically illustrates onset and offset detection using these limits.
10. A script for use in Spike2 for Windows (Cambridge Electronic Design, Cambridge, UK) has been written for 'off-line' analysis of CSP duration. User adjusted settings determine the criteria to be used to detect MEP and silent period signals. These include length of pre- and post-stimulus EMG windows, critical MEP and silent period duration, the number of consecutive data points outside the limit required to recognize signal onset. After rectifying the data, the analysis algorithm finds the mean and MCD of the pre-stimulus EMG data calculating successive differences from data points spaced 0.5 ms apart, independent of the sampling rate. The program then computes the upper and lower limits. Using the limits and the decision criteria, the script recognizes MEP signals where the data exceeds the upper limit and CSP signals where the data is smaller than the lower limit. The results recorded for each are: onset and offset times, duration, latency and area under the curve (and in the case of MEPs, the amplitude). Data in ASCII format can be imported into Spike2 software and analyzed using this script.

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